

LONGITUDINAL DIAGNOSIS DATA



VHA CDW:
nationwide EHR
data from 1,200+
medical centers and clinics



ICD codes
mapped to clinical
text descriptions

data format:

<visit 1>
Hypertensio,
Type 2 diabetes mellitus with
diabetic chronic kidney disease,
</visit 1>

<visit n>
Chronic kidney disease,
Chronic low back pain,
</visit n>

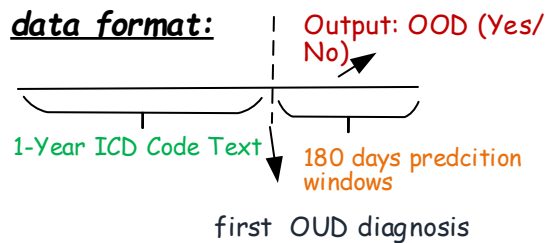
OOD COHORT

OOD fine-tuning cohort



Outcome cohort
78,660 VHA patients
with OUD 62,928 train
· 7,866 val · 7,866 test

data format:



External transfer cohort



MIMIC-IV: 3,971
patients with OUD
55 opioid overdose
events within 180 days

CONTINUED PRETRAINING and FINE TUNING



Continued pretraining on Large-
scale longitudinal ICD-text
sequences
3,984,788 VHA patients

ClinicalMamba-2.8B



OOMamba
output: binary logits (0/1)

Qwen3-1.7B



OODQwen
output: binary logits (0/1)

MULTI-EXPERT FRAMEWORK

OODQwen trained
Qwen3-4B trained
Qwen3-8B

Fusion Router
Combine expert logits

Final OOD Prediction

Fusion strategies

EW:
Equal-
Weight
Fusion

PGA:
Prior-
Guided
Adaptive
Fusion

GLA:
Global-
Local
Adaptive
Fusion